

Junhyun An

Statistics & Mathematics, Seoul National University | anjunhyun@snu.ac.kr | 82-10-2415-7065

<https://anjunhyun.github.io/anjunhyun/>

Education

Seoul National University, BS in Statistics and Mathematical Sciences

Mar 2021 – Present

Research Interests

Knot Theory • Graph Theory • Algebraic & Geometric Topology

Research Experience

Research & Education (R&E)

May 2019 - Dec 2019

Advisor: Prof. Gyotaek Jin

The Smallest Lattice Cube Containing Nontrivial Band Knots

- Developed Python algorithms for generating all band knots in an $n \times n \times n$ lattice cube.
- Proved that every band knot in a $3 \times 3 \times 3$ lattice cube is ambient isotopic to the unknot.
- Established the existence of a Möbius strip in $2 \times 2 \times 2$ and a trefoil knot in $4 \times 4 \times 4$.

Independent Research

May 2020 - Dec 2020

Feb 2023 - Sep 2023

Computational Complexity and Approximation of the Rural Postman Problem in Euclidean Graphs

- Proved NP-hardness by showing that Euclidean TSP is a special case.
- Developed a polynomial-time upper bound using minimum matching, doubled edges, and a minimum spanning tree.

Undergraduate Research Internship

Dec 2024 – Dec 2025

Advisor: Prof. Jisu Kim

Periodicity testing for time series(Student-Directed Education Program)

Dec 2024 – Dec 2025

- Co-developed a topological method for testing periodicity in time series.
- Proved asymptotically valid confidence bounds for one-dimensional homology features.
- Validated the method on non-sinusoidal periodic signals through simulations.

Random Graph Models for Knot Generation

Dec 2024 – Jul 2025

- Developed computational pipelines to study knot distributions arising from random graphs embedded in various manifolds.
- Identified limitations of statistical inference caused by the sparse occurrence of nontrivial knots.

Persistent Homology of Subsampled Paley Graphs

Jul 2025 – Dec 2025

- Constructed weighted filtrations on subsampled Paley graphs over finite fields.
- Observed correlations between persistent homology and Gauss sum magnitudes, including distinctions between prime and prime-power fields.

Academic Activities

Knot Theory Seminar(Student-Directed Education Program)

Mar. 2026 – Jun. 2026

- Designed the overall seminar curriculum by selecting lecture topics and organizing a coherent progression.
- Co-developed lecture slides and delivered presentations covering core topics in classical knot theory.
- Created exercises and solution sessions to reinforce key concepts and motivate subsequent topics.

Preprint

- [1] D. Park, J. An, T. Kim, and J. Kim. “Subsampling Confidence Bound for Persistence Diagrams via Time-delay Embedding.” *arXiv:2512.06324*, 2025. (Co-first author)